ACQUITY UPLC Systems with 2D Technology

Capabilities Guide

Revision A

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Overview

Waters ACQUITY UPLC Systems with 2D Technology employ solvent managers (pumps), a sample manager, a column manager, optical detectors, mass spectrometers, and a wide range of separation chemistries to facilitate multidimensional analyses. To meet the specific needs of your application, you can configure a 2D system with as many as three ACQUITY UPLC solvent managers and as many as three 515 pumps (or one 1525 pump). The column manager (CM-A), which is a required part of all 2D systems, utilizes programmable switching valves, and two, independent heating zones with active preheating. Various core systems, all in the UPLC family, are available as foundations for 2D technology.

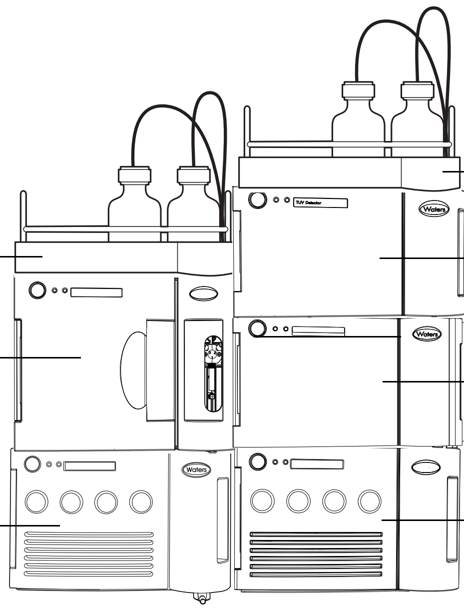
Tip: You can upgrade a conventional ACQUITY UPLC system to include 2D technology.

The features of a system with 2D technology make these goals achievable:

- Eliminate matrix effects
- Improve peak capacity (gain selectivity and sensitivity)
- Improve ruggedness (cleaner MS source, extended column lifetime)
- Obtain higher quality data for decision making

Components and Features of an ACQUITY UPLC System with 2D Technology

An ACQUITY UPLC system with 2D Technology:



A general example of an ACQUITY UPLC system with 2D Technology is shown in the illustration, above. The following specific items are required components of the system:

- A column manager (CM-A), with programmable switching valves
- Waters Pump Control
- Windows 7 or Windows XP operating systems. (See the appropriate MassLynx SCN release notes for the required workstation configuration.)
- MassLynx version 4.1 with one of the following software change notes (SCN): SCN 781, SCN 802, SCN 805, and SCN 810.
- ACQUITY UPLC Systems June 2011 Driver Pack.

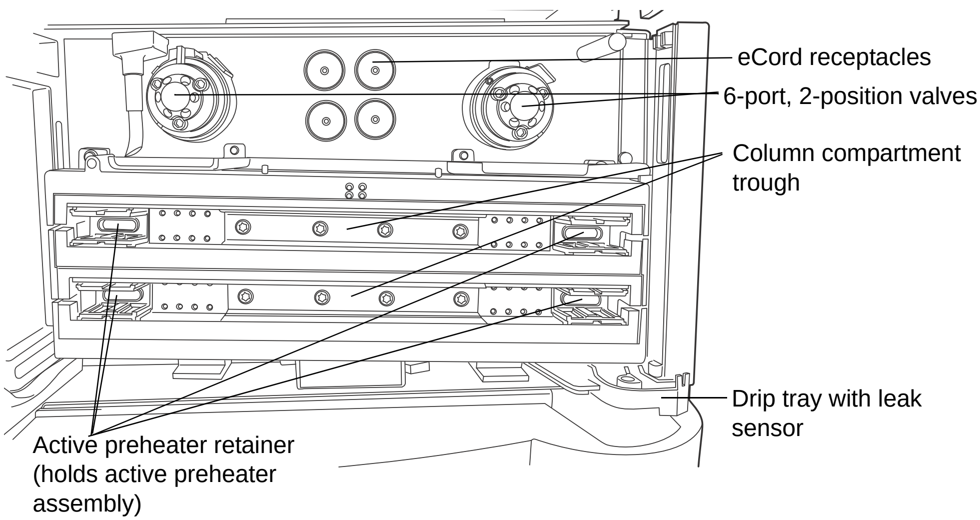
Column manager (CM-A)

The column manager (CM-A) includes two 6-port, 2 position valves and two column troughs with independent temperature control. You can use a maximum of two active pre-heaters when running 2D applications in the CM-A. In the event table, you specify the valve (Left Valve or Right Valve) and position of each at a specific time.

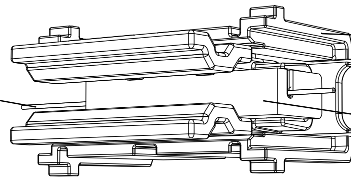
Requirement: Confirm that each column is at the desired temperature before running samples. It is possible to begin an injection before a column reaches its temperature set point when operating an ACQUITY UPLC System with 2D Technology.

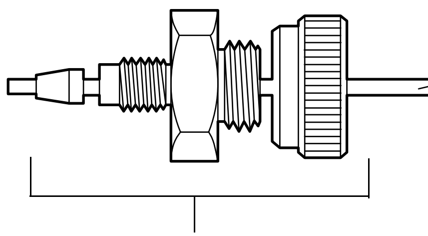
Restriction: Auxiliary column heaters (CM-Aux) are not supported with 2D systems.

Column manager with door open:



Active preheater assembly:



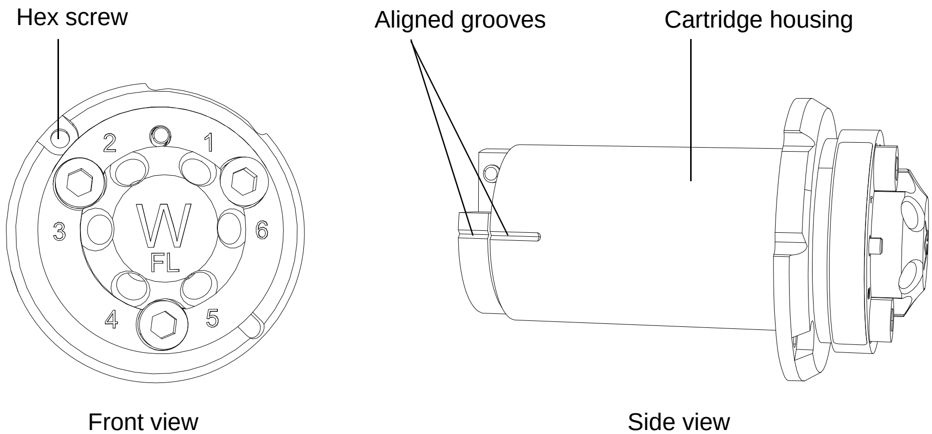


Column manager components:

Component	Description
Active preheater assembly	Heats the incoming solvent to the specified column temperature before it enters the column.
Active preheater retainer	Secures the active preheater in the column compartment.
Valves	6-port, 2-position valve
Drip tray	Captures any leakage from the column or column connections and routes it to the drip management system.

See also: *ACQUITY UPLC Column Compartments Operator's Overview and Maintenance Information.*

Valve (6-port, 2-position)



CM-A Valve Cartridge P/N	Description
205000952	Kit, ACQUITY UPLC 2D Technology SS Valve
205000953	Kit, ACQ UPLC 2D Tech. Bio-Inert Valve

Tip: Bio-inert valves are intended for use with highly aqueous, highly ionic, mobile phases.

Solvent manager

Typical system configurations include two binary solvent managers, or one quaternary solvent manager and one binary solvent manager. ACQUITY UPLC Systems with 2D Technology can utilize as many as three ACQUITY UPLC solvent managers.

Requirement: The quaternary solvent manager must be configured as the first-dimension pump (Pump 1) in the Waters Pump Control software (see page 12).

Caution: To avoid problems with retention time reproducibility, do not use a quaternary solvent manager for the second dimension of your analysis.

See also: Your system's user documentation for additional information on solvent managers.

Additional pumps

You can use as many as three 515 pumps, or one 1525 pump, in a 2D system. These additional pumps are often used for at-column dilution, or for adding a reagent. The additional pumps can achieve a flow rate of up to 10 mL of solvent per minute, but their 5000 psi pressure limit usually governs their pumping rate. Both the 1525 pump and the pump control module (PCM) that controls the 515 pumps communicate with the system's workstation via IEEE 488.

See also:

- *Waters 1525µ and 1525EF HPLC Pump Installation and Maintenance Guide*
- *Waters Pump Control Module II Installation Guide*

External valves

The CM-A includes two, 6-port, 2-position valves, and can also independently control as many as three (3) external valves (listed below) through your analysis method. The external valves have separate power supplies and communicate with the CM-A through an RS232 connection (via cable, P/N 441000459).

Important: If you intend to upgrade an existing CM-A as part of a new 2D system that includes external valves, consult your local service engineer, to ensure compatibility.

External Valve P/N	Ports	Positions	Material	Pressure Rating
417000138	10	2	Titanium	5,000 PSI
417000139	7	6	PEEK	5,000 PSI
417000150	11	10	Stainless Steel	5,000 PSI

Tubing

High-pressure, critically clean, tubing assemblies are available for use with 2D systems. You can choose stainless-steel or MP35N tubing assemblies, depending on your application. Use MP35N for corrosion resistance in high-ionic strength aqueous conditions, such as those needed for separating biological samples.

See also:

- Spare parts information on page 35 for more information on tubing assemblies.
- Waters Spare Parts Locator on waters.com for a current list of tubing assemblies.

Configuring the system

Important:

- To enable 2D features, you must operate the CM-A in Advanced mode.
- For information on pump and sample manager configurations other than those listed in this document, contact Waters.
- When operating any system that includes multiple pumps and valves, you must know the pressure limitations of all system components including flow cells, pumps, valves, and columns. Moreover, you must consider pressures at all stages of the method during system operation, mindful that configurations and operations vary in the level of pressure they produce within a system and that the pressure on individual components within systems can vary.

Caution: To avoid communication problems when you add or replace an instrument component of an ACQUITY UPLC System with 2D Technology, repeat the procedure for configuring the system, below. Proper communication cannot be achieved unless you repeat the configuration procedure for the new instrument.

To configure the system:

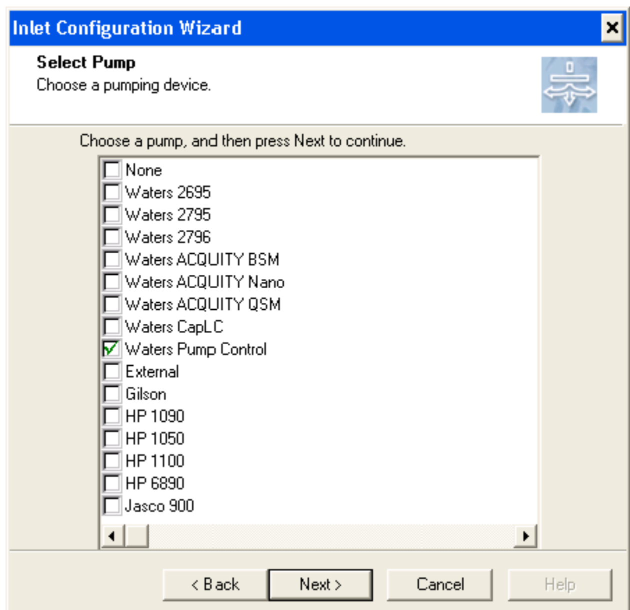
1. Verify that all necessary instrument cables are connected and that all instruments are powered on.

2. Open MassLynx, and select Inlet Method.

3. Select Tools > Instrument Configuration, to open the Inlet Configuration dialog box.

4. Click Configure, to open the wizard and then click Next on the wizard Welcome page.

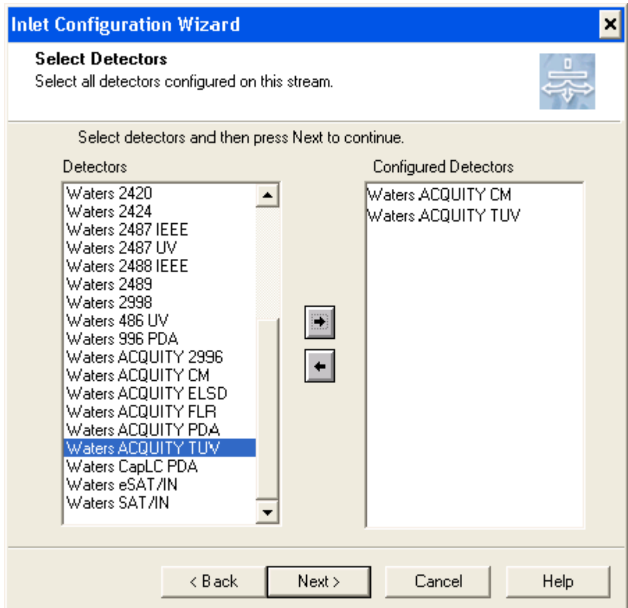
5. On the Select Pump page, choose Waters Pump Control, and then click Next.



6. On the Select Auto Samples page, select the applicable Sample Manager, and then click Next.

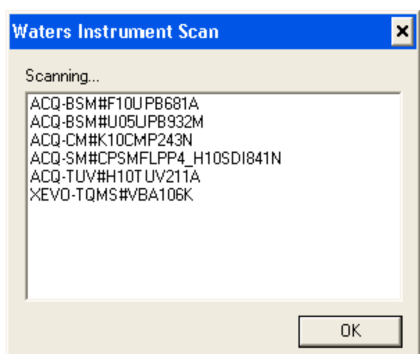
7. On the Select Detectors page select Waters ACQUITY CM, and use the right-hand arrow to drag the selected detector to the Configured Detectors area.

8. Sequentially select each applicable detector and drag it to the Configured Detectors area, and then click Next.

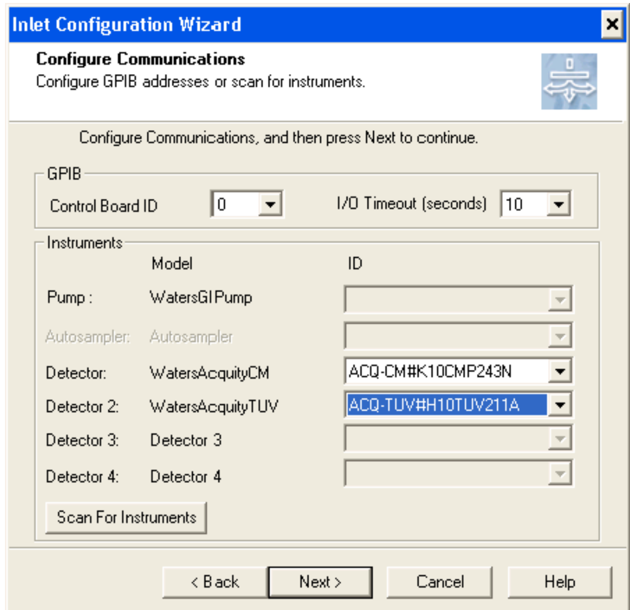


9. On the Configured Communications page, click Scan for Instruments.

10. Verify the list of ACQUITY modules that appears in the Waters Instrument Scan dialog box is correct, and then click OK.



11. Select the instrument ID from each menu, and then click Next.



12. When the Configuration Successful prompt appears, click Finish.

13. If any of the detectors (including the column manager) do not appear in the Selected Configuration Addresses section, repeat step 4 through step 12 before proceeding to step 14.

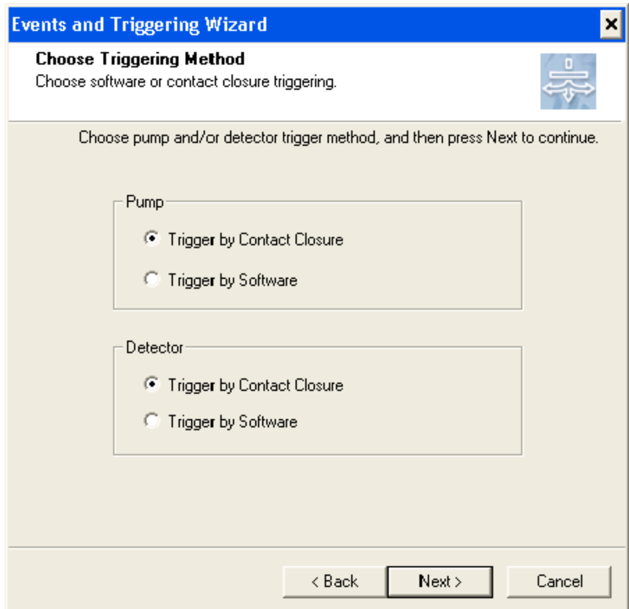
14. From the Inlet Configuration page, click Events and Triggering, and then click Next.

15. On the Choose Events page, click Next, leaving the default settings unchanged.

    Note: You do not need to configure event wires.

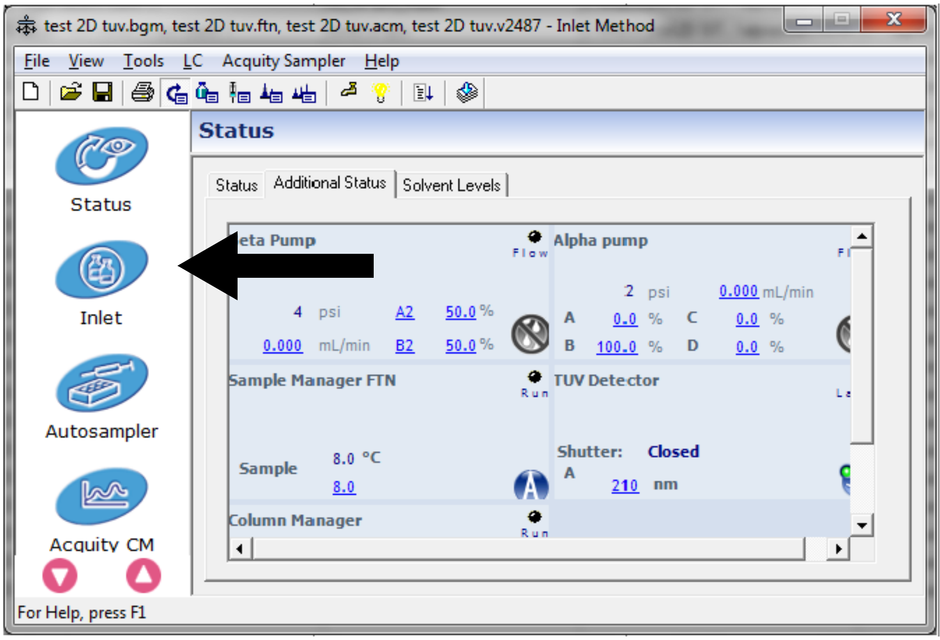
16. On the Choose Triggering Method page, select Trigger by Contact Closure for the Pump and Detector, click Next, and then click Finish.

Important: Do not select Trigger by Software. For additional information, refer to PCS Number 22273 in the Known Issues table (see page 30).

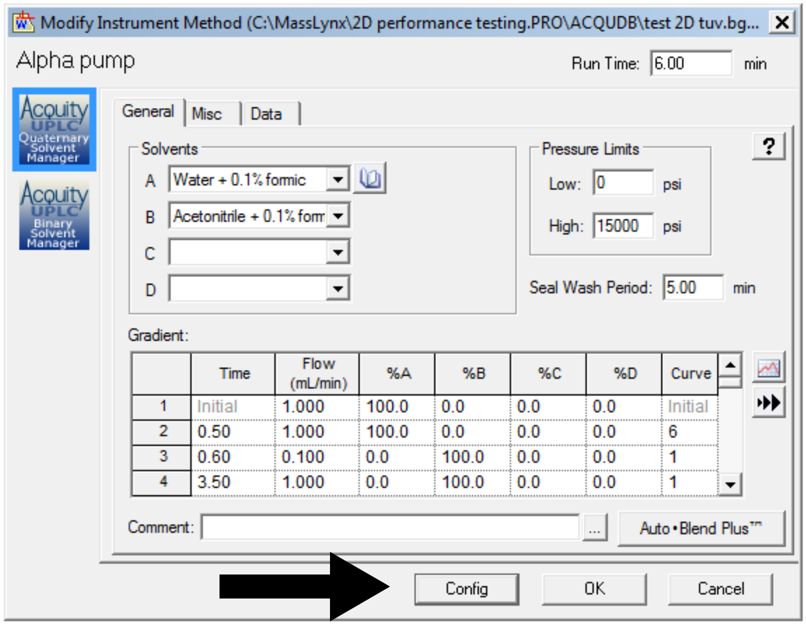


To configure Waters Pump Control:

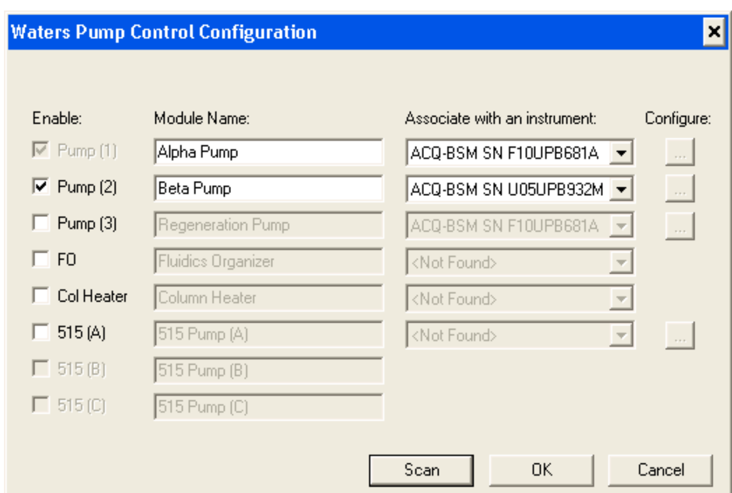
1. From the Inlet Editor, click Inlet.



2. In the instrument method window, click Config.

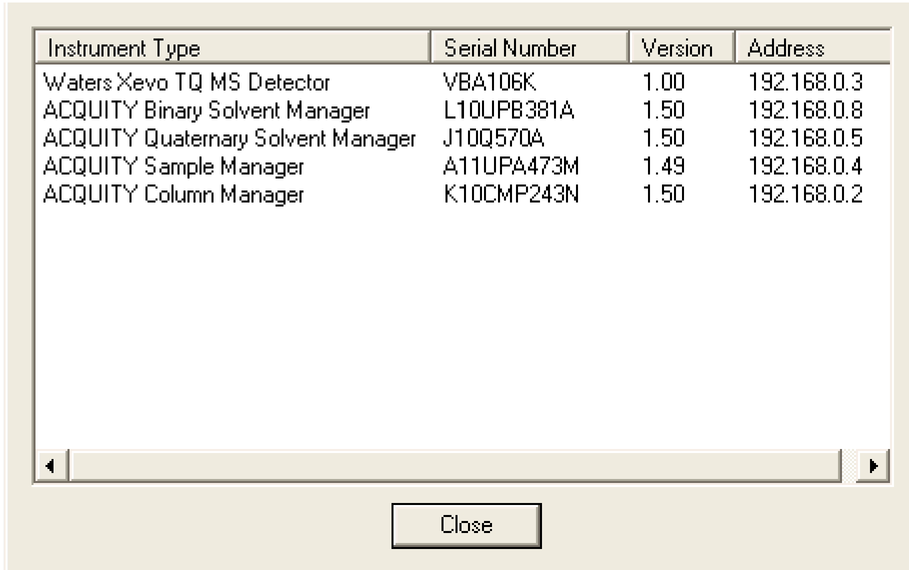


3. Enter an appropriate module name for each pump. The example, shown below, uses “Alpha Pump” for Pump 1, and “Beta Pump” for Pump 2.



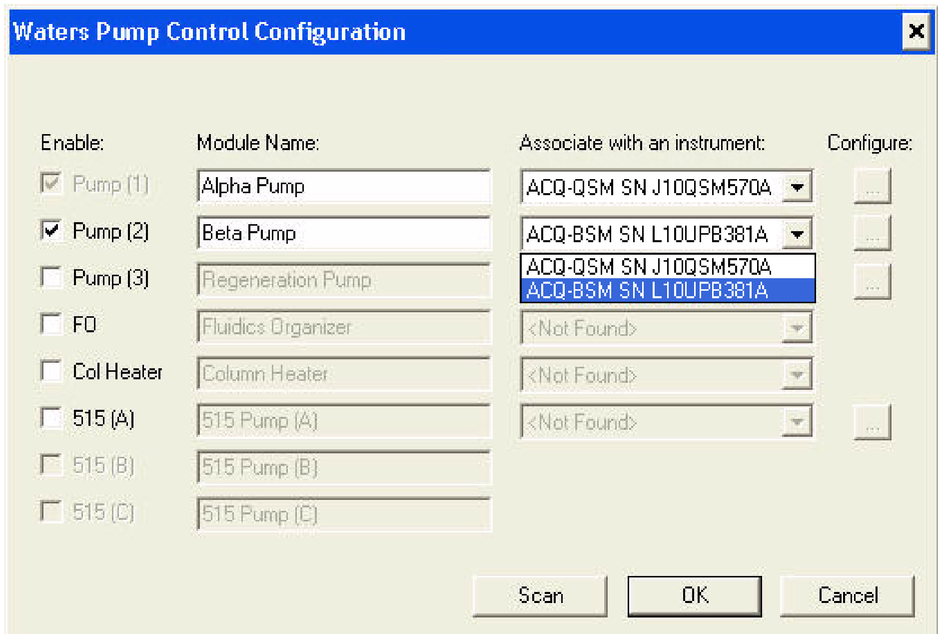
4. Click Scan.

5. After confirming the presence of all instruments and devices, click Close.



Instrument Type	Serial Number	Version	Address
Waters Xevo TQ MS Detector	VBA106K	1.00	192.168.0.3
ACQUITY Binary Solvent Manager	L10UPB381A	1.50	192.168.0.8
ACQUITY Quaternary Solvent Manager	J10Q570A	1.50	192.168.0.5
ACQUITY Sample Manager	A11UPA473M	1.49	192.168.0.4
ACQUITY Column Manager	K10CMP243N	1.50	192.168.0.2

6. On the Waters Pump Control Configuration page, from the “Associate with an instrument” menus, select the appropriate pumps and associated serial numbers.



7. Click OK.

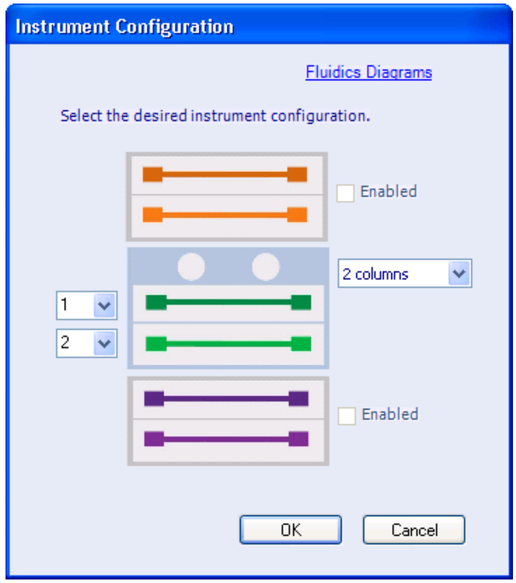
8. When prompted, click OK, to confirm and reset communications.

To configure the CM-A:

1. From the Column Manager in ACQUITY Console software, click Configure > Instrument.

2. In the Instrument Configuration Dialog box, from the menu on the right-hand side, select two columns.

Requirement: Ensure that the column position selections on the left-hand side are set as shown, below.

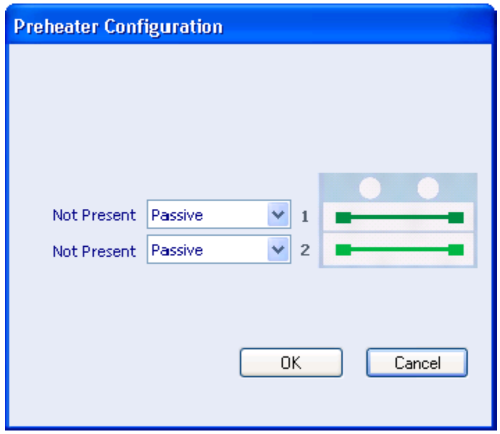


Restriction: Do not select more than two columns.

3. Click OK to accept the configuration.
4. From the Column Manager in ACQUITY Console, select Configure > Operating Mode.

   Result: The Operating Mode page appears.
5. Click Advanced, and then select 2 for Right and Left positions, and 6 for Right and Left ports, and then click OK.
6. When the Configuration Changed dialog box appears, cycle the power to the CM-A, and then click OK, to restart and open the ACQUITY Console software.
7. Reset Communications in the Inlet Method Editor.
8. From the Column Manager in the ACQUITY Console, click Configure > Preheaters.

Result: The Preheater Configuration page appears.



Restriction: You can install only one pre-heater in each trough, on either the left-hand or right-hand side. On the pre-heater configuration page, select the desired pre-heater configuration for each trough.

9. Select Active or Passive for column positions 1 and 2.

10. Click OK.

11. When the Configuration Changed dialog box appears, cycle power on the CM-A, then click OK to restart the ACQUITY Console software.

    See also: *ACQUITY UPLC Column Compartments Operator's Overview and Maintenance Information* for more details on the CM-A.

Connecting external valves to the CM-A

Requirement: If you intend to upgrade an existing CM-A as part of a new 2D system, consult your local service engineer, to ensure compatibility. For more information, see PCS Number 22194 in the Known Issues table (see page 30).

To connect an external valve:

1. Attach the adapter cable (P/N 441000459) to the connector labeled COM B on the rear panel of the CM-A.

2. Attach one end of the serial cable supplied with the valve to the adapter cable and the other end to the valve.

3. Using the address dial on the rear of each valve, set the appropriate address.

   Requirement: Each valve must have a unique address setting: 1, 2, or 3, that corresponds to the external valve identification: 1, 2, or 3.

4. Power-on the CM-A.

5. Power-on the external valves.

   Note: See PCS Number 22194 in the Known Issues table (see page 30) when initializing the valves.

To configure the valves:

1. In the ACQUITY Console software, select the column manager.
2. Click Configure/Operating System.
3. Mark the check box for the external valve(s) you are using.

Example system configurations

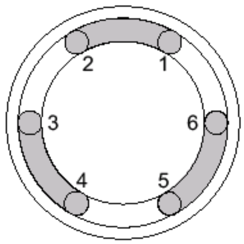
ACQUITY UPLC Systems with 2D Technology are designed to provide maximum flexibility when configuring flow paths for your applications.

Some possible plumbing configurations are listed, below:

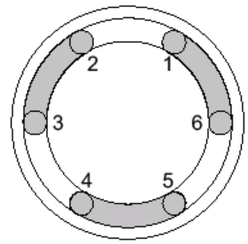
- Trap and back-transfer
- Filter trapping
- Heart-cutting
- Parallel column regeneration
- At-column dilution

Important: During installation, a Waters service engineer runs a system installation test for a basic “trap and back-transfer” configuration to ensure that the instrument modules are performing optimally and that the fluid path is configured correctly.

Valve positions



Valve
Position 1



Valve
Position 2

Valve usage

Typical usage conditions for the trap and back-transfer configuration are a flow rate of 1.0 mL/min for Pump 1 running water and 0.7 mL/min running 50:50 acetonitrile/water for Pump 2. Column 1 is a 2.1 × 30 mm column with a 20 mm particle size and Column 2 is a 2.1 × 50 mm column with a 1.7 mm particle size. Pressures of 6895 kPa (68.95 bar, 1,000 psi) are seen by the alpha pump and pressures of 62050 kPa (621 bar, 9,000 psi) are seen by the beta pump.

Recommendation: Replace valves after 50,000 cycles of typical use.

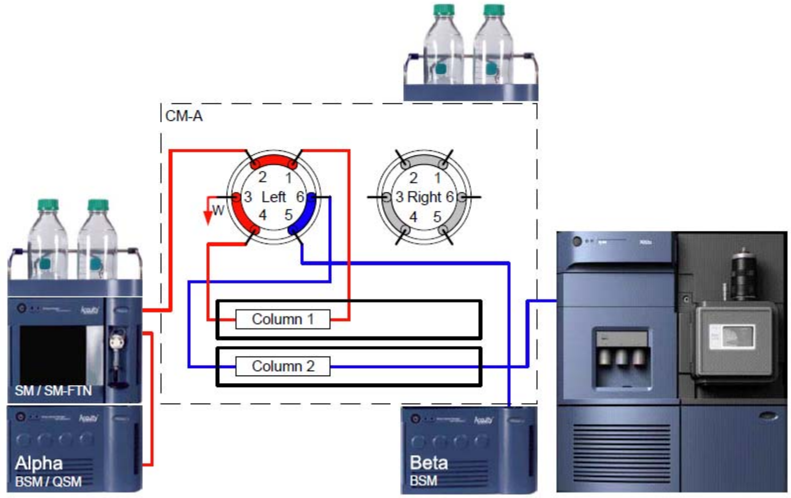
Trap and back-transfer configuration

Note: In the following example, Pump 1 (first dimension) is labeled “Alpha” and Pump 2 (second dimension) is labeled “Beta”.

Trapping is when analytes of interest are captured on a trapping column (first dimension) and then transferred to a second column for an analytical separation (second dimension).

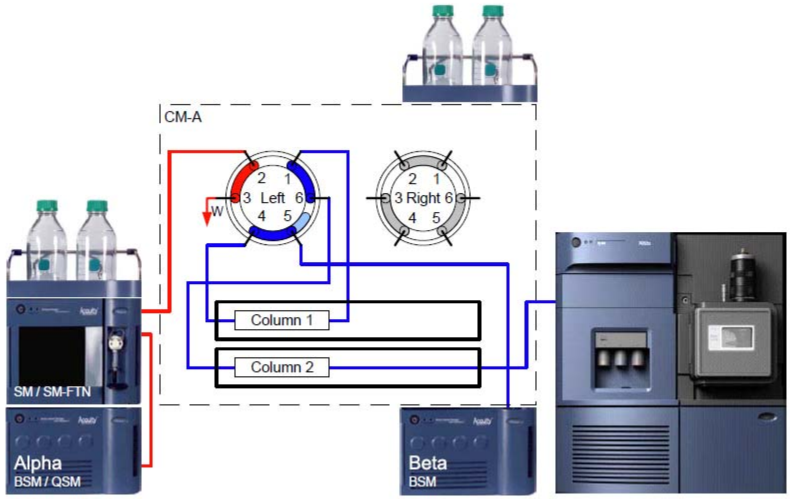
You can use a trap and back-transfer configuration for a variety of applications including online sample preparation or to simplify your separation.

Trap and back-transfer step 1:



In this configuration, the alpha pump loads the sample onto column 1, trapping analytes of interest. The alpha pump then performs a wash that retains analytes of interest and allows some impurities to flow to waste. After the wash, the alpha pump decreases its flow rate and the left-hand valve in the CM-A switches from position 1 to position 2.

Trap and back-transfer step 2:



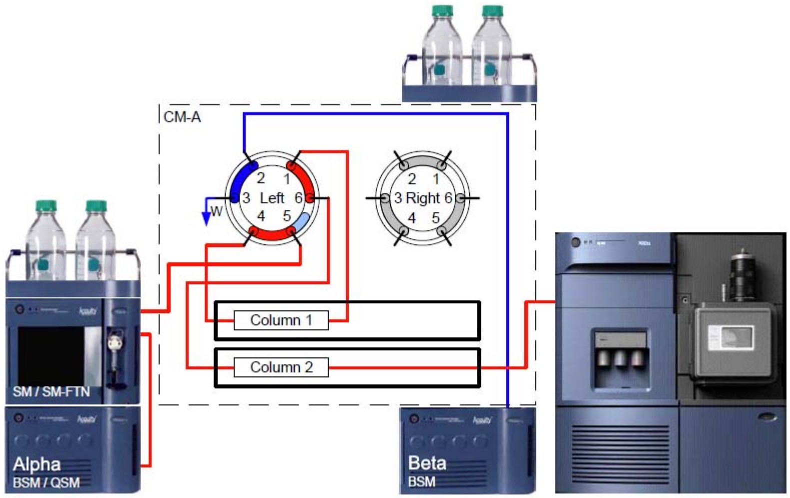
With the left-hand valve in position 2, the beta pump passes a gradient through column 1, transferring the analytes to column 2 for the analytical separation.

Filter trapping configuration

Note: In the following example, Pump 1 (first dimension) is labeled “Alpha” and Pump 2 (second dimension) is labeled “Beta”.

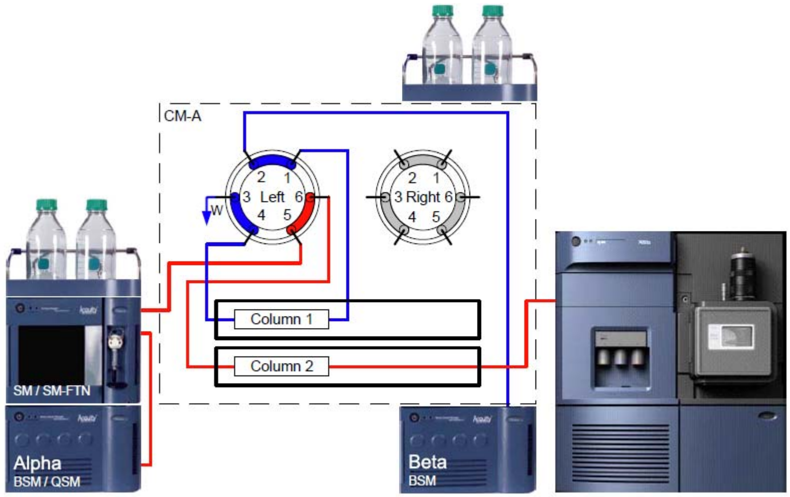
The filter trapping configuration prevents impurities from reaching the head of the analytical column.

Filter trapping step 1:



The sample is injected using flow from the alpha pump. The analytes of interest pass through column 1, which acts as a filter and traps impurities. When the analytes have passed through column 1 they are focused onto the head of column 2. The alpha pump then decreases flow and the left-hand CM-A valve switches from position 2 to position 1.

Filter trapping step 2:



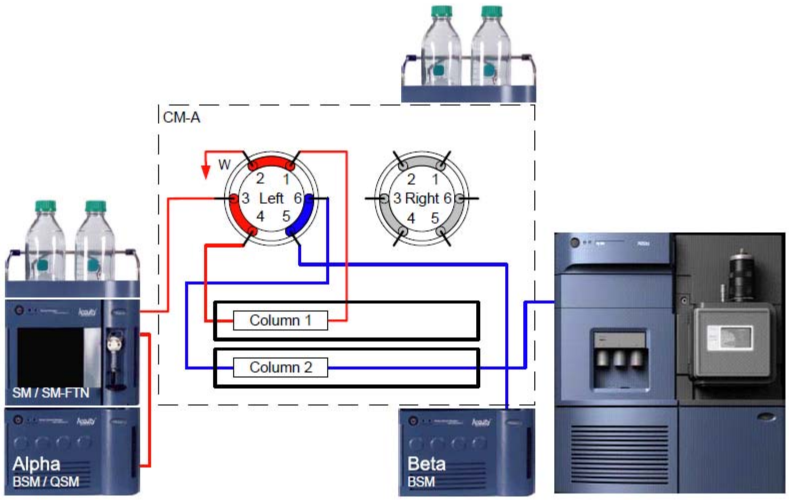
When the left-hand CM-A valve switches from position 2 to position 1, the beta pump starts flow to recondition column 1 and send any impurities to waste. The alpha pump starts flow through column 2 to perform the analytical separation.

Heart-cutting configuration

Note: In the following example, Pump 1 (first dimension) is labeled “Alpha” and Pump 2 (second dimension) is labeled “Beta”.

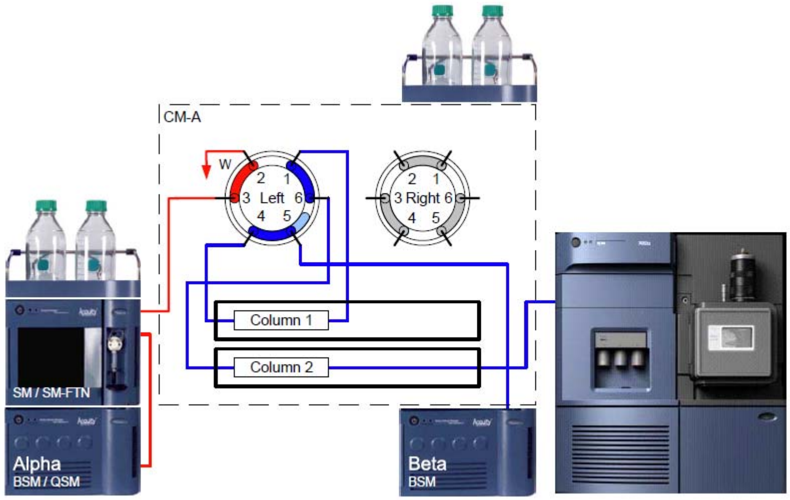
Heart-cutting is a function whereby analytes of interest elute from the first analytical column and are then diverted onto a second column for another analytical separation. Use heart-cutting for obtaining alternate selectivity, or for reducing sample load for better resolution. The following example details one method of heart-cutting.

Heart-cutting step 1:



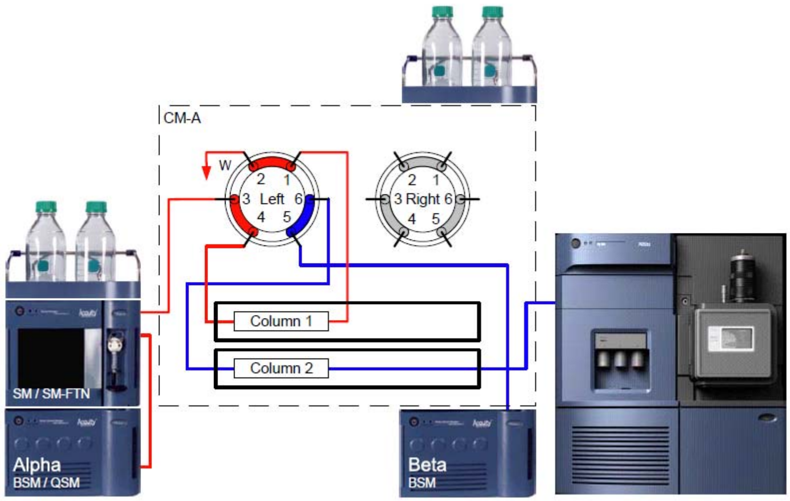
The alpha pump performs an analytical separation on column 1. A particular predefined segment of that chromatogram is diverted for a second analytical separation. As the predefined segment of the analytical separation elutes from column 1, the left-hand CM-A valve switches from position 1 to position 2.

Heart-cutting step 2:



With the left-hand CM-A valve in position 2, the beta pump elutes analytes of interest from column 1 and diverts them to column 2. When the segment of the analytical separation on column 1 is diverted to column 2, the left-hand CM-A valve switches from position 1 to position 2.

Heart-cutting step 3:



With the left-hand CM-A valve in position 1, the beta pump performs an analytical separation of the diverted segment on column 2. The alpha pump re-equilibrates column 1.

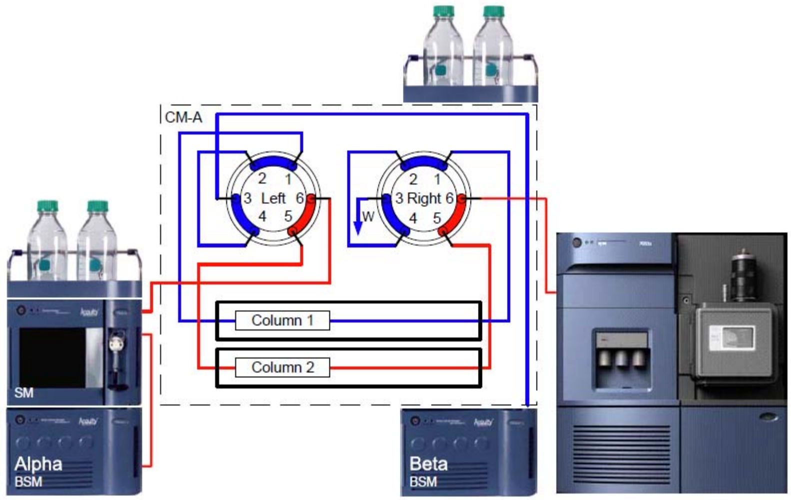
Parallel column regeneration configuration

Note: In the following example, Pump 1 (first dimension) is labeled “Alpha” and Pump 2 (second dimension) is labeled “Beta”.

In parallel column regeneration, one column is used for analysis, while a second column is re-equilibrated, preparing for the next injection. Over a series of samples and analytical separations, these columns alternate.

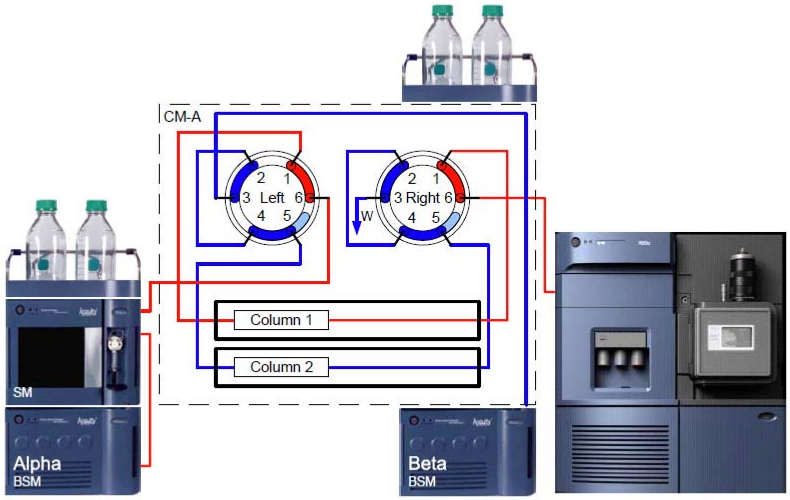
Parallel column regeneration is useful for increasing throughput.

Parallel column regeneration step 1:



An injection, using flow from the alpha pump, moves the sample onto column 2 for the analytical separation. The beta pump re-equilibrates column 1, preparing it for the next injection. After the analytical separation is performed on column 2 and column 1 is re-equilibrated, the left-hand and the right-hand CM-A valves switch from position 1 to position 2.

Parallel column regeneration step 2:



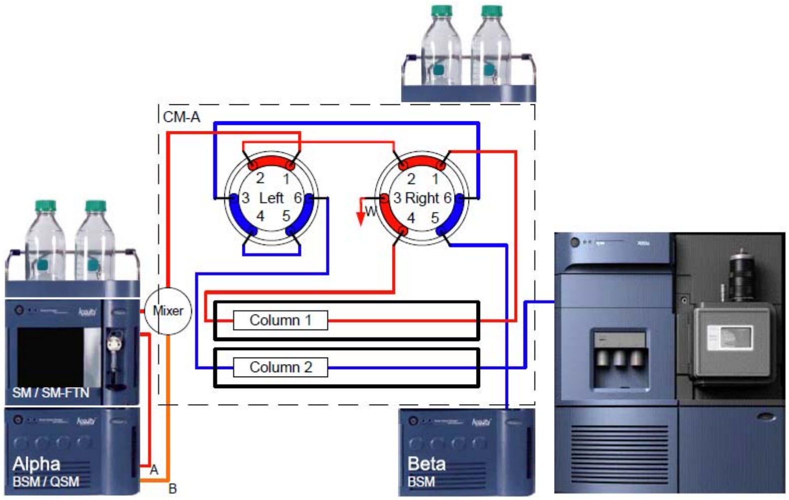
With both the left-hand and right-hand CM-A valves in position 2, the alpha pump provides flow for an analytical separation on column 1 and the beta pump re-equilibrates column 2.

At-column dilution (ACD) configuration

Note: In the following example, Pump 1 (first dimension) is labeled “Alpha” and Pump 2 (second dimension) is labeled “Beta”.

At-column dilution enables direct injection of a large-volume sample in strong solvent by diluting the injected sample before it reaches the head of the column. The following example describes ACD in a trap and back-transfer application.

At-column dilution step 1:

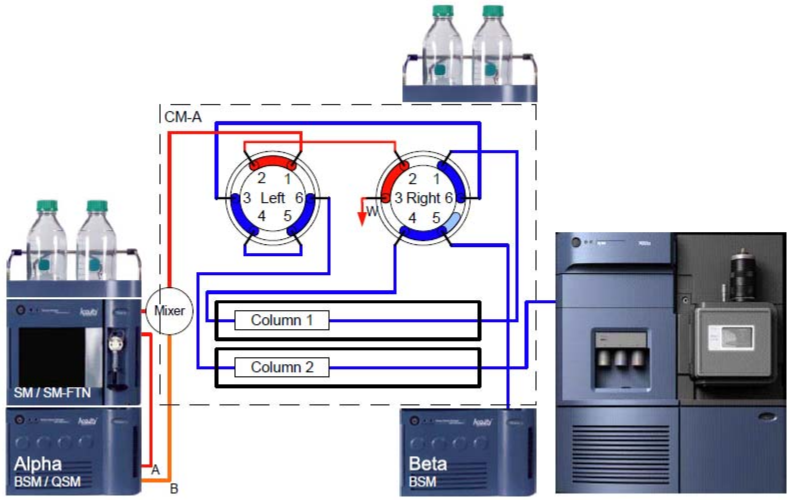


Line A of the loading alpha pump flows organic solvent and pushes the sample into the low-volume mixer (P/N 205000272). Line B of the alpha pump flows 100% aqueous solution into the low-volume mixer, diluting the sample before it reaches column 1.

With both the left-hand and right-hand CM-A valve in position 1, the alpha pump loads the sample onto column 1.

The analytes of interest are then trapped on the head of column 1. The alpha pump then stops flow and the right-hand CM-A valve switches from position 1 to position 2.

At-column dilution step 2:



With the right-hand CM-A valve in position 2, the beta pump starts flow and elutes the analytes of interest from column 1 using a gradient elution onto the more retentive column 2, where the analytical separation is performed.

Known issues

The following table contains a list of the known software issues and workarounds for ACQUITY UPLC Systems with 2D Technology. The PCS (product change system) numbers identify software issues that Waters personnel monitor within a system change request tracking tool.

Module	PCS Number	Description
CM-A	23589, 23762	Following reboot of the console after a CM-A configuration change, columns can appear to be missing from the CM-A tree. **Recommendation:** Restart the CM-A, and then the Console software.
CM-A	23993	Clicking Column QC or Batch QC multiple times while in reading mode can cause the Console to stop functioning properly.

Module	PCS Number	Description
Detectors	22319	If you are using two identical ACQUITY PDA or TUV detectors in an ACQUITY UPLC with 2D Technology system, you can use only one inlet method. The inlet method applies to both detectors. **Tips:** Ensure that both detectors have the same firmware version. Moreover, in the Inlet Configuration wizard, when configuring communications for the two identical detectors, ensure that you give each detector a unique ID.
CM-A	22184	If you are using the advanced mode of the column manager (CM-A), verify that each column is at the desired temperature before running samples. It is possible to begin an injection before a column reaches its temperature set point. **Tip:** You might need to wait for each column to reach the temperature set point.
CM-A	24442	When an 11-port, 10-position valve (P/N 417000150) is used as an external valve, it does not go to position ten when set via the Console software or an instrument method. **Recommendation:** Do not use position 10.

Module	PCS Number	Description
CM-A	22194	External valves connected to early production column managers do not properly initialize. If the column manager has a serial number prior to C11CMP817N or the column manager is part of an existing system that is being upgraded, contact your Waters service engineer to enable the serial port. Every time the early model CM is turned on, power cycled, or the configuration is changed (either in the software or hardware), the unit will need to be turned on and allowed to initialize before power is applied to the external valves. This power-on sequence only applies to column managers that are controlling external valves.
System	22474	To ensure proper operation, you must complete the Inlet Configuration wizard twice, to successfully configure detector addresses. If you do not complete the wizard twice, selected configuration addresses for detectors are not established and do not appear in the Inlet Configuration window. **Requirement:** Click Inlet Method > Tools > Instrument Configuration > Configure. Use the Inlet Configuration Wizard to configure the components of your system. In the Configure Communications window, click Scan For Instruments, and then select IDs for your detectors. Click Finish, to exit the wizard. Launch the Inlet Configuration wizard again, and repeat the steps to configure your system and detectors.

Module	PCS Number	Description
System	24576	The MassLynx sample queue does not always start properly and may hang without error in the “waiting for inlet start” state. **Recommendation:** To recover, select “Stop flow”, to generate an error, reset communications, download the method and restart the sample queue.
System	24501	Waters Pump Control with Masslynx 4.1 generates an error when a pump method (file extension .bgm) size greater than 16k bytes is used. The error message contains: Pump Method Load Method - GetChannels failed with error... The error message also contains the path to the pump method file location.  **Recommendation:** If the size of the pump method file can’t be reduced by eliminating lines from either the gradient table or event table, contact your local Waters representative.

Module	PCS Number	Description
System	22273	To ensure proper operation you must configure your ACQUITY UPLC with 2D Technology system to use contact closures as the trigger method for the pump and detector. You need not do any additional wiring for contact closure triggering of ACQUITY pumps and detectors. Do not configure the system to use software triggers. **Requirement:** In the main MassLynx window, click Inlet Method > Tools > Instrument Configuration > Events and Triggering. In the Events and Triggering Wizard, select Trigger by Contact Closure as the trigger method for the pump and detector.
System	24576	Using the Inlet Prerun and Inlet Postrun functions of the MassLynx sample queue can cause a 2D system to stop functioning properly. **Workaround:** Do not use the Inlet Prerun and Inlet Postrun functions. Instead, use the inlet method editor to run blank injections that create the desired prerun and postrun conditions.

Spare Parts

For information on replacement parts, see the Waters Quality Parts Locator on the Waters web site's Services & Support page.

Part Number	Material	Inside Diameter inch (mm)	Length inch (mm)	Fittings (see note, below)
430001486	Stainless Steel	.010(.254)	24(610)	HP/HP
430002582 700005476	Stainless Steel	.004(.102)	14.5(368)	HP/HP
430002599 700005479	Stainless Steel	.004(.102)	19.0(483)	HP/HP
430002600	MP35N	.004(.102)	14.5(368)	HP/HP
430002641	MP35N	.004(.102)	19.0(483)	HP/HP
430002657 700005480	Stainless Steel	.004(.102)	22.5(572)	LP/LP
430002658 700005485	MP35N	.004(.102)	22.5(572)	LP/LP
430002681 700005477	Stainless Steel	.004(.102)	14.5(368)	LP/LP
430002682 700005482	MP35N	.004(.102)	14.5(368)	LP/LP
430002683 700005478	Stainless Steel	.004(.102)	19.0(483)	LP/LP
430002684 700005483	MP35N	.004(.102)	19.0(483)	LP/LP
430002717	Stainless Steel	.004(.102)	22.5(572)	HP/HP
430002718	MP35N	.004(.102)	22.5(572)	HP/HP
430002727	Stainless Steel	.010(.254)	30(762)	HP/None

Part Number	Material	Inside Diameter inch (mm)	Length inch (mm)	Fittings (see note, below)
430002738	Stainless Steel	.007(.178)	19.0(483)	HP/HP
430002739	Stainless Steel	.004(.102)	27.0(686)	HP/HP
430002769 700005532	Stainless Steel	.004(.102)	36.0(914)	HP/HP
430002772	Stainless Steel	.004(.102)	44.0(1118)	LP/LP
430002779	MP35N	.005(.127)	17(432)	HP/HP
430002852 700005535	Stainless Steel	.004(.102)	22.5(572)	HP/LP
430002972	MP35N	.004(.102)	6.0(152)	HP/HP
430002974	MP35N	.004(.102)	27.0(686)	HP/HP
430002977	MP35N	.004(.102)	36.0(914)	HP/HP
430002980	MP35N	.004(.102)	44.0(1118)	HP/HP
430002983	Stainless Steel	.004(.102)	6.0(152)	HP/HP
430002986	Stainless Steel	.004(.102)	36.0(914)	HP/HP
430002992	Stainless Steel	.004(.102)	44.0(1118)	HP/HP
430002996	Stainless Steel	.007(.178)	14.5(368)	HP/HP
430002999	Stainless Steel	.007(.178)	22.5(572)	HP/HP
430003002	Stainless Steel	.007(.178)	27.0(686)	HP/HP
430003005	Stainless Steel	.007(.178)	36.0(914)	HP/HP
430003008	Stainless Steel	.007(.178)	6.0(152)	HP/HP

Notes:

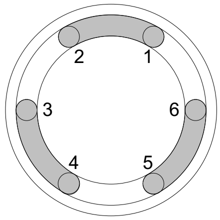
- HP: High-pressure fitting, gold-plated compression screw and two-piece ferrule.
- LP: Low-pressure fitting, single piece PEEK fitting.

Worksheets

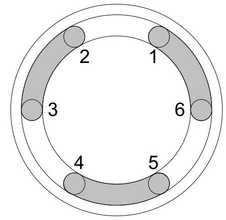
Print the diagrams and tables, below, as needed to document your column manager (CM-A) setup and valve timed events. Both blank worksheets and completed example worksheets are shown.

2D Programming Worksheet

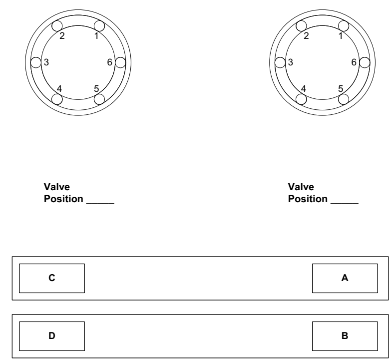
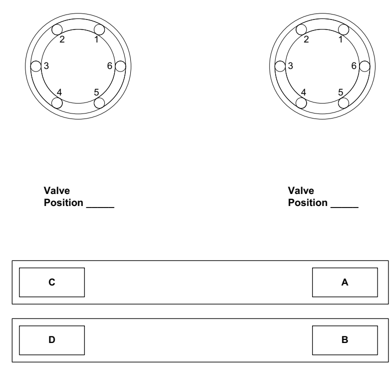
Time	Alpha Pump				Beta Pump				CM-A	
	Flow Rate	%A	%B	Curve	Flow Rate	%A	%B	Curve	Valve	Position

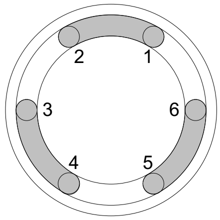


Valve
Position 1

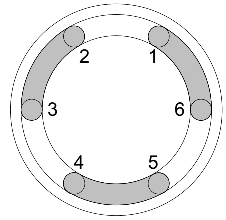


Valve
Position 2

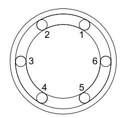
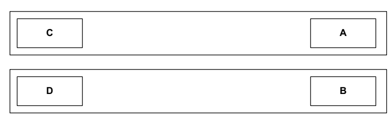
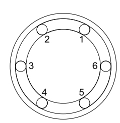
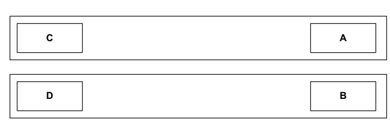
Time	Action	Connections and Position
		
		



Valve
Position 1



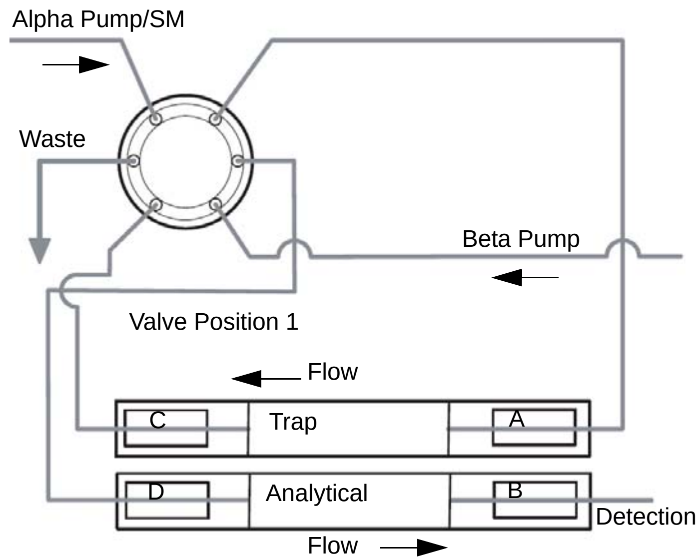
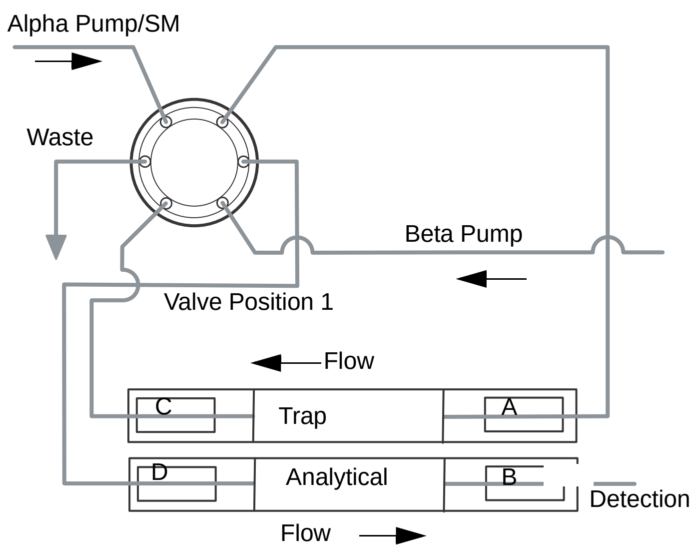
Valve
Position 2

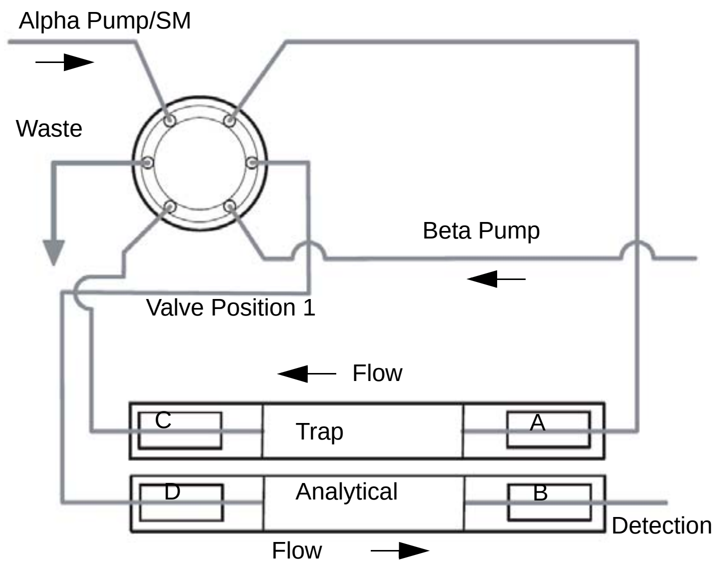
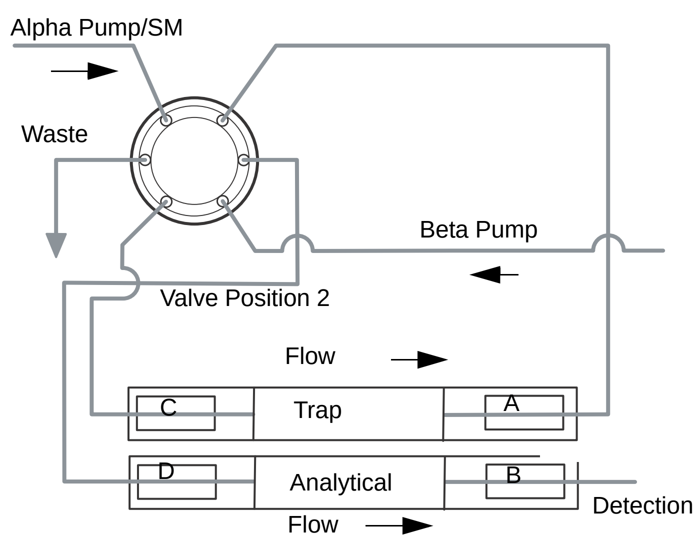
Time	Action	Connections and Position
		 Valve Position _____ 
		 Valve Position _____ 

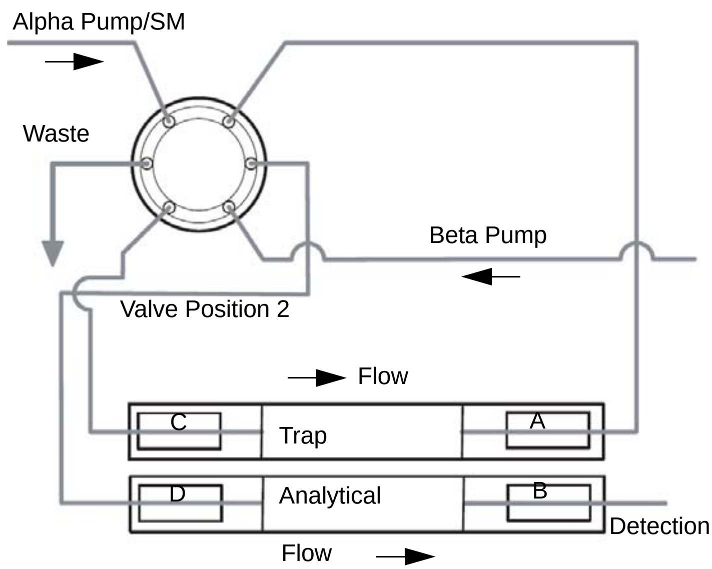
Completed examples

2D Programming Worksheet

	Alpha Pump				Beta Pump				CM-A	
Time	Flow Rate	%A	%B	Curve	Flow Rate	%A	%B	Curve	Valve	Position
0.0	1.0	100	0		0.5	90	10		Left	1
0.5	1.0	100	0	11	0.5	90	10	6	Left	2
0.6	0.1	0	100	11	0.5	90	10	6	Left	2
1.0	0.1	0	100	11	0.5	90	10	6	Left	2
2.0	0.1	0	100	11	0.5	5	95	6	Left	2
3.0	0.1	0	100	11	0.5	5	95	6	Left	2
3.5	1.0	100	0	11	0.5	90	10	6	Left	2
5.5	1.0	100	0	11	0.5	90	10	11	Left	1

Time	Action	Connections and Position
0.0 to 0.5	Inject sample	
0.0 to 0.5	Trap compound	

Time	Action	Connections and Position
0.0 to 0.5	Wash away contaminants	
0.5	Switch valve. Beta pump now flows through the trap column in the reverse direction, transferring analyte to the head of the analytical column.	

Time	Action	Connections and Position
0.5 to 3.5	Analytical separation	
3.5 to 5.5	Switch valve, re-equilibrate system